



Daffodil *International* **University**

Semester Project On -
Computer Networks Lab
CIS211L

Semester Spring-2025

Submitted to-

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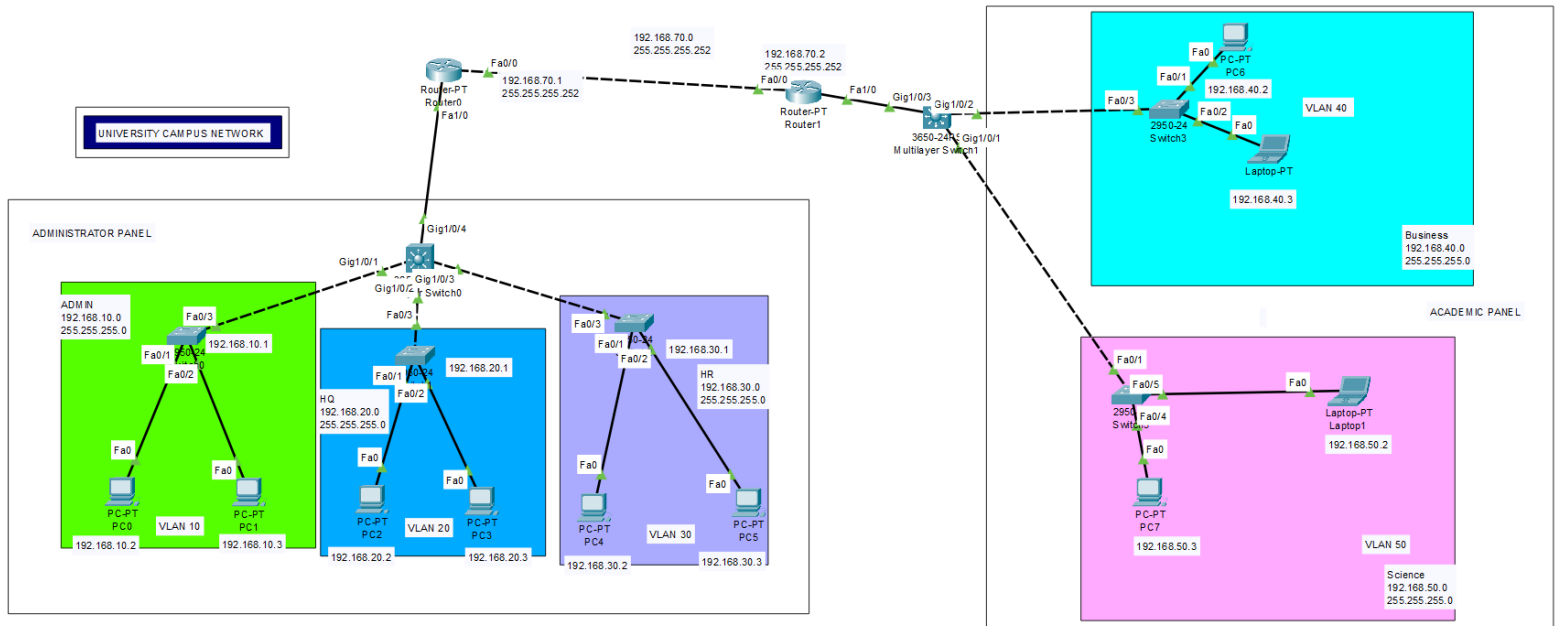
Batch: 18th-(D)

Semester: Spring-2025

Department of Computing and Information System

Project –

In this project, we designed, configured, and simulated a complete network solution for the network of the University Campus using Cisco Packet Tracer.



Here is the visual representation of my project –

Here are a total of 6 networks & 5 VLANs representing 5 Department of University Campus -

Central network of University Campus –

IP-192.168.70.0 Subnet Mask -255.255.255.252

Network of Admin Department – (VLAN 10)

IP-192.168.10.0 Subnet Mask -255.255.255.0

Network of HQ Department – (VLAN 20)

IP-192.168.20.0 Subnet Mask -255.255.255.0

Network of HR Department – (VLAN 30)

IP-192.168.30.0 Subnet Mask -255.255.255.0

Network of Business Department – (VLAN 40)

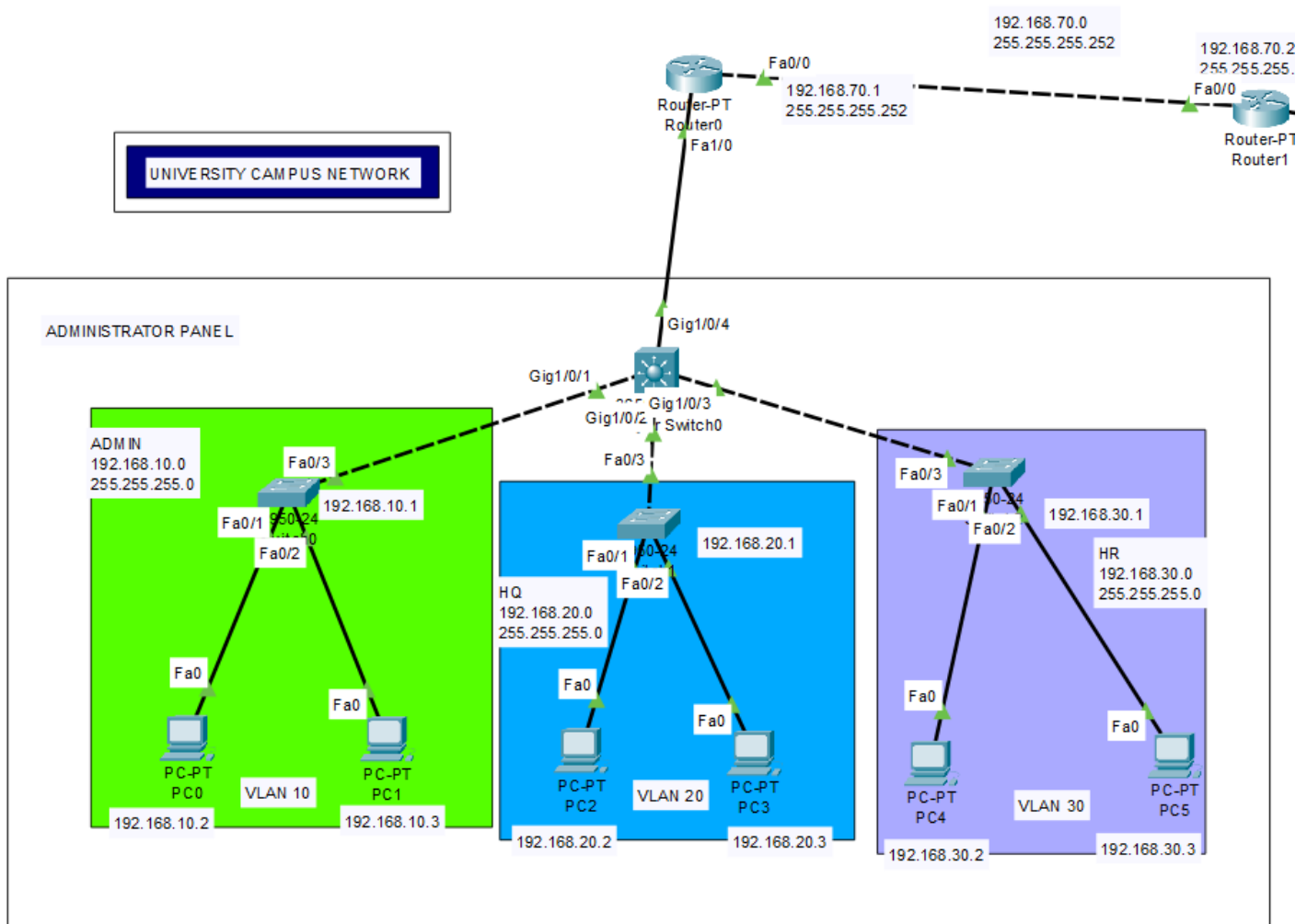
IP-192.168.40.0 Subnet Mask -255.255.255.0

Network of Science Department – (VLAN 50)

IP-192.168.50.0 Subnet Mask -255.255.255.0

Here, my network has 2 main panels.

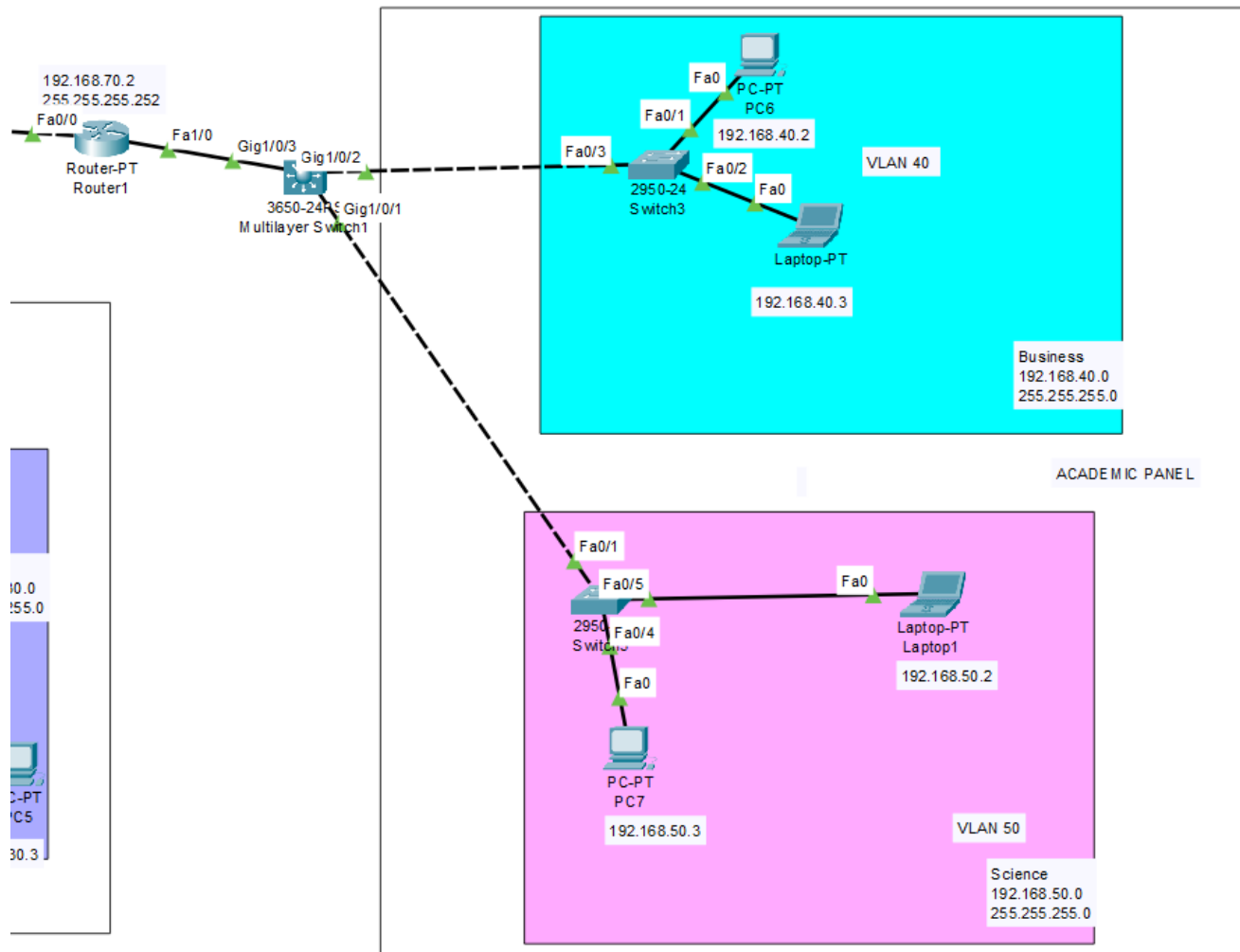
One of them is the Administrator Panel connected to Router -0, and under this panel, we have 3 departments: Admin, HQ, and HR. They all have their own VLAN corresponding: VLAN 10, VLAN 20, VLAN 30.



But in this panel, all departments can contact each other as in an organization, departments need to contact each other to operate the administrator panel, and also they can contact the departments of another panel.

Realtime Simulation										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC4	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC3	Laptop0	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC2	PC7	ICMP		0.000	N	3	(edit)	(delete)

The other Panel is the Academic Panel connected to Router-1, where I have 2 departments, Business & Science. They have also their own corresponding VLANs 40 & 50.

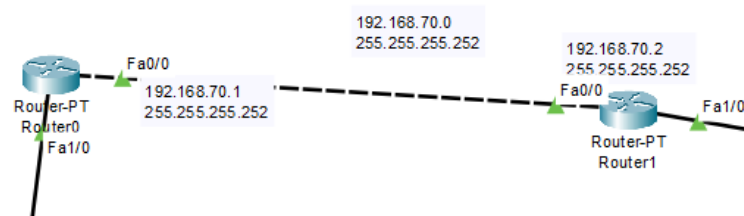


But unlike the other one in this panel they can't contact with each other due to their privacy regarding their faculty and students info. But they can contact with the Administrator Panel if needed.

Realtime Simulation										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC6	PC7	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC4	Laptop0	ICMP		0.000	N	1	(edit)	(delete)
	Failed	PC7	Laptop0	ICMP		0.000	N	2	(edit)	(delete)
	Successful	Laptop1	PC3	ICMP		0.000	N	3	(edit)	(delete)

I used both of the routers in the network as a DHCP Server, responsible for assigning IP addresses to client devices dynamically.

I have used RIP version 2 dynamic routing protocols to manage routing between networks established between 2 routers.



1st we configured router physically in usual manner then we drew a network and assign them IP & subnet mask shown above.

1. Here is the detailed routing table –

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router-0	Fa1/0.10	192.168.10.1	255.255.255.0	-
Router-0	Fa1/0.20	192.168.20.1	255.255.255.0	-
Router-0	Fa1/0.30	192.168.30.1	255.255.255.0	-
Router-0	Fa0/0	192.168.70.1	255.255.255.252	-
PC0 (Admin)	Fa0/1	DHCP	DHCP	DHCP
PC1 (Admin)	Fa0/2	DHCP	DHCP	DHCP
PC2 (HQ)	Fa0/1	DHCP	DHCP	DHCP
PC3 (HQ)	Fa0/2	DHCP	DHCP	DHCP
PC4 (HR)	Fa0/1	DHCP	DHCP	DHCP
PC5 (HR)	Fa0/2	DHCP	DHCP	DHCP
Router-1	Fa1/0.40	192.168.40.1	255.255.255.0	-
Router-1	Fa1/0.50	192.168.50.1	255.255.255.0	-
Router-1	Fa0/0	192.168.70.2	255.255.255.252	-
PC6 (Business)	Fa0	DHCP	DHCP	DHCP

Laptop0(Business)	Fa0	DHCP	DHCP	DHCP
PC7 (Science)	Fa0	DHCP	DHCP	DHCP
Laptop1 (Science)	Fa0	DHCP	DHCP	DHCP

Here is the detailed CLI configuration code for my project –

For the network of Administrator Panel:

Switch-Admin:

```

en
conf t
vlan 10
name Admin
exit
int fa0/1
switchport access vlan 10
exit
int fa0/2
switchport access vlan 10
exit
int fa0/3
switchport mode trunk
exit
interface range fa0/1-2
switchport mode access
exit
do wr

```

Switch-HQ:

```

en
conf t
vlan 20
name HQ
exit

```

```
int fa0/1
switchport access vlan 20
exit
int fa0/2
switchport access vlan 20
exit
int fa0/3
switchport mode trunk
exit
interface range fa0/1-2
switchport mode access
exit
do wr
```

Switch-HR:

```
en
conf t
vlan 30
name HR
exit
int fa0/1
switchport access vlan 30
exit
int fa0/2
switchport access vlan 30
exit
int fa0/3
switchport mode trunk
exit
interface range fa0/1-2
switchport mode access
exit
do wr
```

For the Multi-Layer-Switch to be the trunk:

```
en
conf t
vlan 10
name Admin
exit
vlan 20
name HQ
exit
vlan 30
name HR
exit
interface range gig1/0/1-4
switchport mode trunk
exit
do wr
```

Configuration the port connected with the core switch on router-0:

```
en
conf t
int fa1/0
no shutdown
exit
```

```
int fa1/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit
```

```
int fa1/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
```

```
int fa1/0.30
encapsulation dot1Q 30
```



```
ip address 192.168.30.1 255.255.255.0
exit
do wr
```

Applying DHCP on router-0:

```
en
conf t
hostname dhcp-server
```

```
ip dhcp pool Admin
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
exit
```

```
ip dhcp pool HQ
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
exit
```

```
ip dhcp pool HR
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
dns-server 8.8.8.8
exit
```

```
ip dhcp excluded-address 192.168.10.1
ip dhcp excluded-address 192.168.20.1
ip dhcp excluded-address 192.168.30.1
do wr
```

For the network of Academic Panel:-

Switch-Business:

```
en
conf t
vlan 40
name Business
exit
int fa0/1
switchport access vlan 40
exit
int fa0/2
switchport access vlan 40
exit
int fa0/3
switchport mode trunk
exit
interface range fa0/1-2
switchport mode access
exit
do wr
```

Switch-Science:

```
en
conf t
vlan 50
name Science
exit
int fa0/4
switchport access vlan 50
exit
int fa0/5
switchport access vlan 50
exit
int fa0/1
switchport mode trunk
```

```
exit
interface range fa0/4-5
switchport mode access
exit
do wr
```

For the MultiLayer-Switch to be the trunk:

```
en
conf t
vlan 40
name Business
exit
vlan 50
name Science
exit
interface range gig1/0/1-3
switchport mode trunk
exit
do wr
```

Configuration the port connected with the core switch on router-1:

```
en
conf t
int fa1/0
no shutdown
exit
```

```
int fa1/0.40
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
exit
```

```
int fa1/0.50
encapsulation dot1Q 50
ip address 192.168.50.1 255.255.255.0
exit
```

```
do wr
```

Applying DHCP on router-1:

```
en
conf t
hostname dhcp-server-baek

ip dhcp pool Business
network 192.168.40.0 255.255.255.0
default-router 192.168.40.1
dns-server 8.8.8.8
exit
```

```
ip dhcp pool Science
network 192.168.50.0 255.255.255.0
default-router 192.168.50.1
dns-server 8.8.8.8
exit
```

```
ip dhcp excluded-address 192.168.40.1
ip dhcp excluded-address 192.168.50.1
do wr
```

Connection Between Router-1 & Router-2 using RIP v2:

In Router-0:

```
en
conf t
int fa0/0
ip address 192.168.70.1 255.255.255.252
no shut
exit
router rip
version 2
network 192.168.70.0
```

```
network 192.168.10.0
network 192.168.20.0
network 192.168.30.0
no auto summary
exit
```

In Router-1:

```
en
conf t
int fa0/0
ip address 192.168.70.2 255.255.255.252
no shut
exit
router rip
version 2
network 192.168.70.0
network 192.168.40.0
network 192.168.50.0
no auto summary
exit
```

Apply ACL in Router-1 to restrict access between VLAN 40 & VLAN 50:

```
en
conf t

access-list 140 deny ip 192.168.40.0 0.0.0.255 192.168.50.0 0.0.0.255
access-list 140 permit ip 192.168.40.0 0.0.0.255 any
interface fa1/0.40
ip access-group 140 in

access-list 150 deny ip 192.168.50.0 0.0.0.255 192.168.40.0 0.0.0.255
access-list 150 permit ip 192.168.50.0 0.0.0.255 any
interface fa1/0.50
ip access-group 150 in

do wr
```

Here is the drive link of my Cisco Packet Tracer and the configuration code –
https://drive.google.com/drive/folders/1xov8_UhzNGn7LXviC-3Rt4AoSFuLtR_b?usp=sharing

```
//*To remove ACL
en
conf t
interface fa1/0.40
no ip access-group 140 in

interface fa1/0.50
no ip access-group 150 in

no access-list 140
no access-list 150*//
```